

Question

A is a binary ionic compound containing the anion X^- , which can be synthesized from the corresponding elemental substances under normal conditions via a combination reaction, with a formula weight not exceeding $80\text{g} \cdot \text{mol}^{-1}$.

The quaternary compound **B** is a hydrate of a strongly alkaline ionic compound, with mass fractions of C, H, and O being 26.51%, 12.79%, and 52.97%, respectively.

When solid **B** reacts with excess solid **A** under vacuum, three products are obtained after extraction and separation with liquid ammonia: a salt **M** containing the anion X^- , a strongly alkaline ionic compound **C**, and a colorless gas **D**. The anion X^- is derived from **D** by gaining electrons.

Determine the sum of the stoichiometric coefficients (in their simplest whole-number ratios) for all substances in the overall reaction equation for the formation of compound **M** from compounds **A** and **B**.

A. 63

B. 4

C. 20

D. 11

E. 12

F. 8

G. None of the above options are correct.

Answer

Correct Answer: **A**

Detailed Explanation

Since compound **A** is a binary ionic compound, it is highly likely that its cation and anion correspond to a single element each. Therefore, X^- contains only one element, and since X^- is obtained by the electron gain of the colorless gas **D**, **D** must be a colorless gaseous elemental substance, either O_2 or N_2 . As X^- carries only a single negative charge, X^- can only be O_2^- or N_3^- , and the cation must be an alkali metal ion.

CHECKPOINT

1 PTS

X^- can only be O_2^- or N_3^-

CHECKPOINT

2 PTS

The cation must be an alkali metal ion

Since compound **A** can be formed under normal conditions by the direct combination of elemental substances, and nitrogen gas cannot directly combine with alkali metals other than Li, nor can it produce azide ions even when combined with Li, **D** must be oxygen, and X^- is the superoxide ion. Sodium metal cannot directly combine with oxygen under normal conditions to form a superoxide, whereas potassium can. The atomic weights of rubidium and cesium would violate the condition of a molar mass less than 80 g/mol. Therefore, compound **A** must be potassium superoxide KO_2 , and **D** is oxygen.

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CHECKPOINT

2 PTS

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CHECKPOINT

1 PTS

Compound **A** is potassium superoxide KO_2

CHECKPOINT

1 PTS

D is oxygen

For compound **B**, the molar ratio of C, H, and O is determined to be 4:23:6 based on the mass fraction.

CHECKPOINT

2 PTS

The molar ratio of C, H, and O in compound **B** is 4:23:6

Assuming 6 oxygen atoms, the molar mass of **B** is $181.23\text{g} \cdot \text{mol}^{-1}$. Subtracting the contributions of C, H, and O leaves 14.01, corresponding exactly to one nitrogen atom. Thus, the chemical formula of **B** is $C_4H_{23}O_6N$.

Considering it is a strongly alkaline ionic compound with hydration, the alkalinity must originate from an anion, and its basicity is less than or equal to that of hydroxide. Based on the chemical formula, it is reasonable to infer that the source of alkalinity is hydroxide.

CHECKPOINT

1 PTS

The alkalinity of **B** must originate from an anion, and its basicity is less than or equal to that of hydroxide

CHECKPOINT

1 PTS

The chemical formula is $C_4H_{23}O_6N$

The remaining formula $C_4H_{22}O_5N$ corresponds to the cation and water of crystallization. No reasonable solution exists for 1, 2, 3, or 4 water molecules. With 5 water molecules, the cation formula is $C_4H_{12}N$, corresponding to the tetramethylammonium ion. Therefore, the chemical formula of **B** is $[Me_4N]^+[OH]^- \cdot 5H_2O$.

CHECKPOINT

1 PTS

The chemical formula of **B** is $[Me_4N]^+[OH]^- \cdot 5H_2O$

Clearly, based on reactivity, KO_2 will react with the water of crystallization in **B** to produce O_2 and hydroxide ions, so hydroxide must appear among the three products. The tetramethylammonium ion of **B** and the potassium ion of **A** must each combine with an anion after the reaction. Since the product **M** contains the superoxide ion, its

cation cannot be potassium (otherwise, no reaction would occur). Therefore, **M** is $[Me_4N]^+[O_2]^-$, and **C** must be KOH.

CHECKPOINT

1 PTS

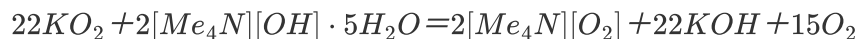
M is $[Me_4N]^+[O_2]^-$

CHECKPOINT

1 PTS

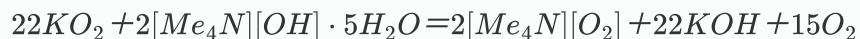
C must be KOH

Balancing the equation yields:



CHECKPOINT

2 PTS



The sum of the coefficients is 63, so the answer is A.

CHECKPOINT

1 PTS

The sum of the coefficients is 63